VIPERFISH

CABLE CONFIGURATION

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Contents

1 Abstract 3

2 Towing characteristics 3

2.1 Fairing 5

2.2 Layback and cable length standard tow cable 6

2.3 Layback and cable length for tow cable with 100 m hair fairing 7

2.4 Layback and cable length for tow cable with 500 m hair fairing (10 kts) 8

3 Cable tension with ViperFish 9

4 Recommended cables 10

1 Abstract

Hydrodynamic analysis of the tow cable has an important role when configuring a ViperFish system. The cable catenary including cable properties, cable depth corresponding to given cable length and cable tension are parameters that affect the ViperFish system. This document assumes a Ø9.2 mm cable, which is the preferred cable size when towing a ViperFish.

The cable is analysed at different payouts and speeds when towing the ViperFish and presents the catenary profile that compares the depth and position after the towing vessel.

2 Towing characteristics

This section includes graphical figures that represent the towing characteristics of the ViperFish for various cable lengths, speeds and depths.

The following ViperFish layback figures are provided to assist in towing the system. Please note the following points if the figures are used as reference:

- The standard ViperFish currently supports operational depths of up to 200 meters.
- The results listed in these tables are calculated using the EIVA NaviSuite Beka cable program with the best-estimated parameters for vehicle weight, drag, lift and buoyancy, as well as cable weight and stiffness. EIVA disclaims any liability for consequential damage resulting from using the data in the tables.
- The values contained in these figures are subject to change and have been assigned an error of +/-10%. Keep this in mind, especially if within 10% of the limits of any parameter.
- These guidelines are to be used with caution until verified by actual sea trials.

The figures contained within this document have been generated under the following assumed conditions:

- Sea state = 0 (no vessel heave motion)
- Solutions are for steady-state tow vessel speeds (2 kts to 10 kts, 2-kt increments)
- The cable used is Ø9.2 mm (faired Ø9.8 mm)
- Cable length 50 m to 1,700 m

Ocean current will be a factor for virtually all cases in which the system is used. By following the simple procedure below, the existing figures can be used for predicting the ViperFish depth when ocean currents are present (assuming the ocean current is uniform with depth).

The speed specifications of the system are measured in speed through water (STW). Therefore, if you are going with the current, you need to increase the vessel speed by the

same amount as the current. For example, for the speed of the ViperFish to be 3 knots STW, and you are sailing with the current which is 2 knots, then you need to have a vessel speed of 5 knots speed over ground (SOG).

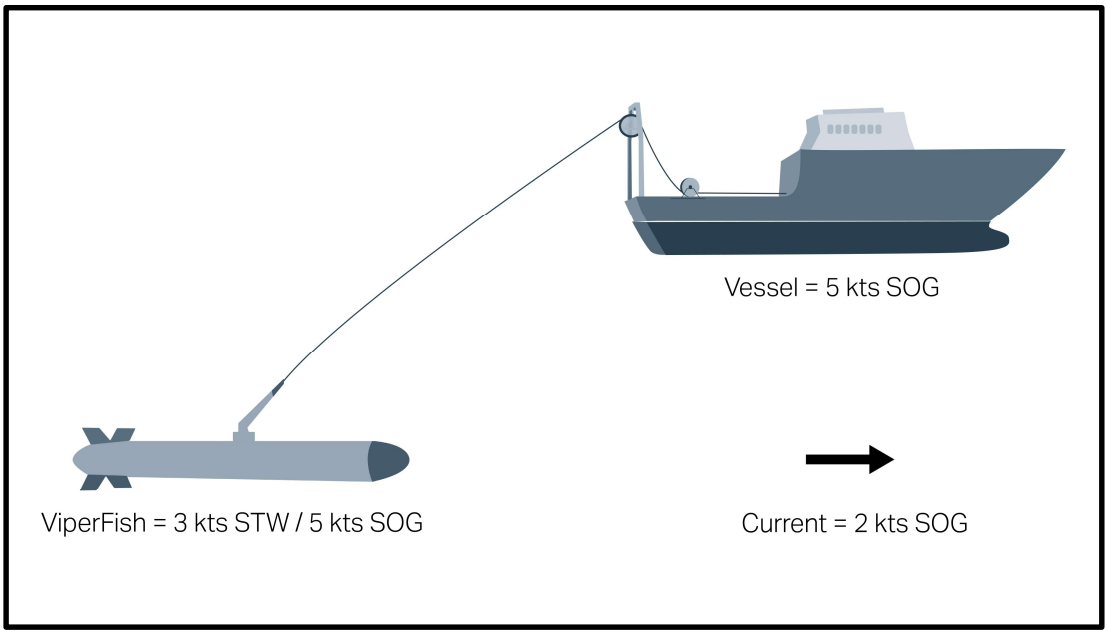

Figure 1 Vessel speed when sailing with the current to achieve a target STW for the ViperFish

And the other way around, as STW speed changes, drag will change accordingly.

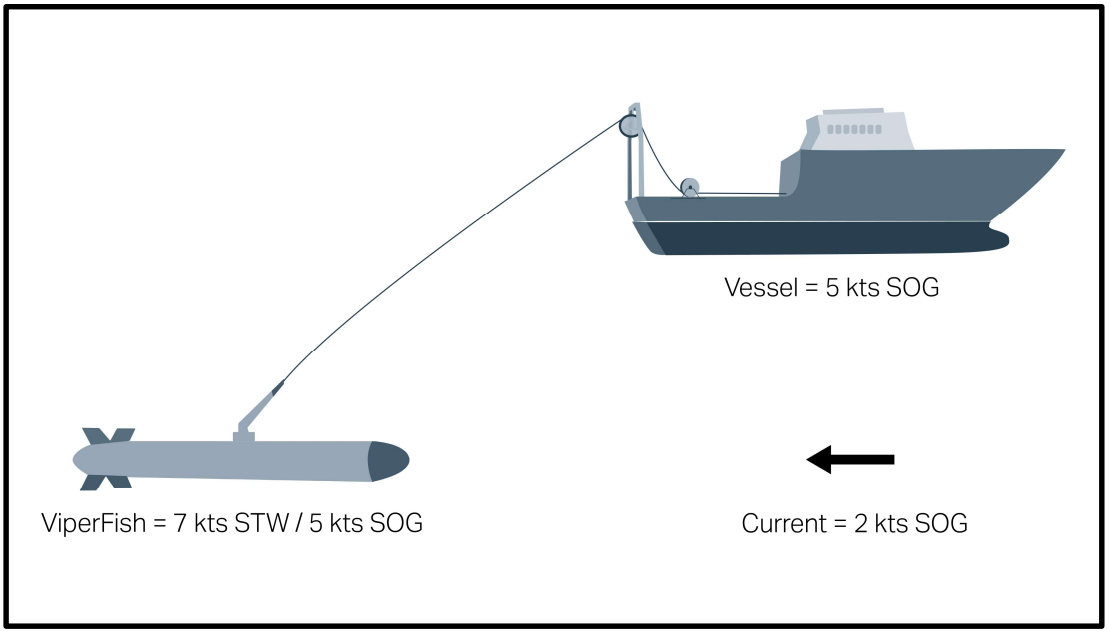

Figure 2 Vessel speed when sailing against the current to achieve a target STW for the ViperFish

2.1 Fairing

A fairing is a structure attached to a towed cable, designed to streamline the flow around the cable. Cables are faired primarily for two reasons:

- To reduce normal drag and thus achieve more depth for a given cable scope and speed
- To eliminate cable vibration caused by vortex shedding, commonly known as cable strum

Hair fairing is a hair like flexible structure attached to cable that assumes a streamlined shape when towed. Drag coefficient Cd: 0.6.

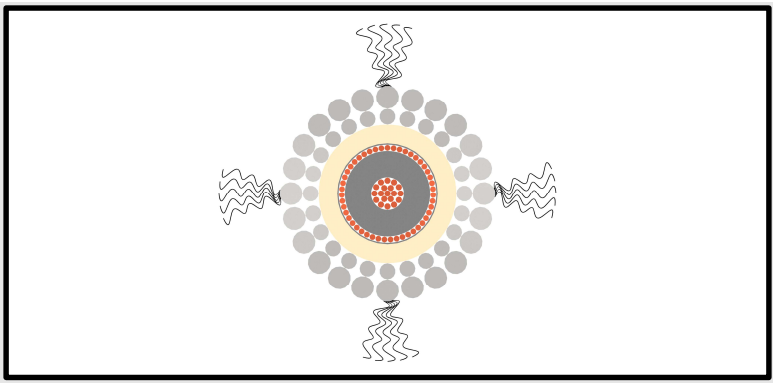

Figure 3 A hair-faired cable

2.2 Layback and cable length standard tow cable

The following graphs show the results for a standard tow cable of Ø9.2 mm with no fairing.

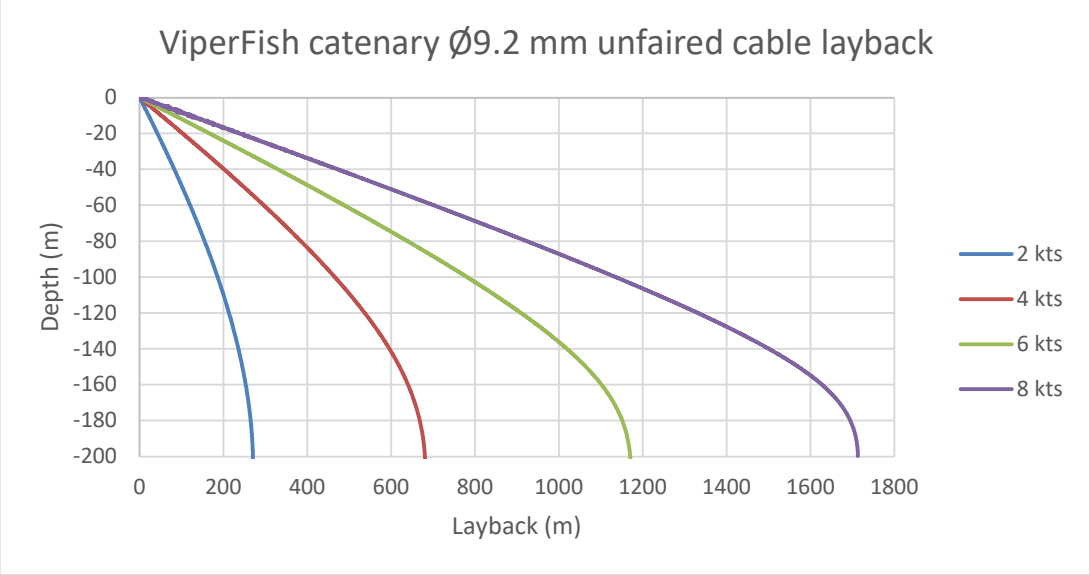

Figure 4 ViperFish cable layback after vessel versus depth (cable with no fairing)

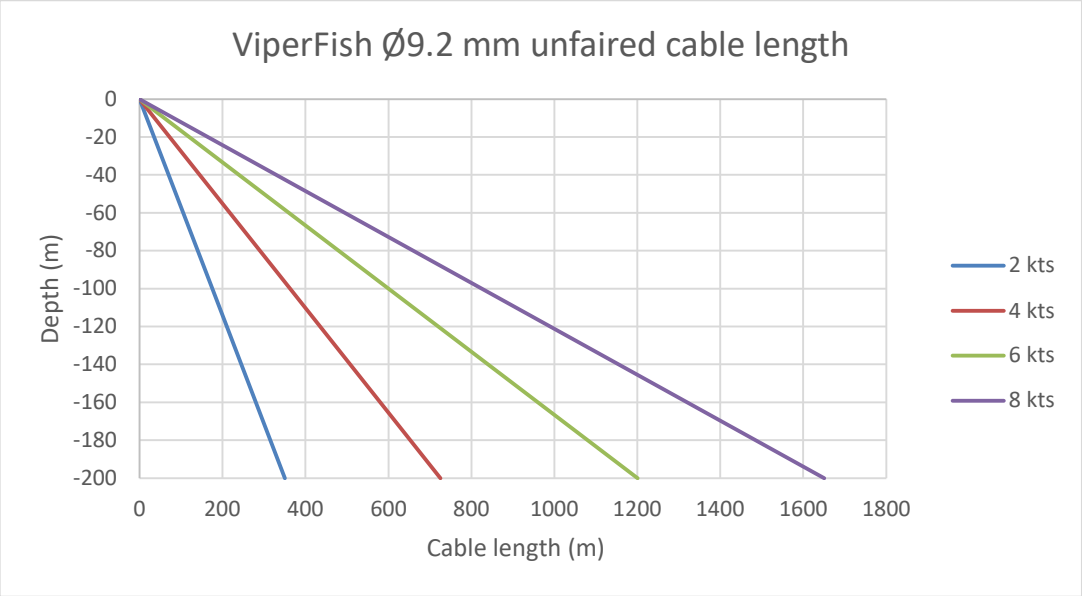

Figure 5 ViperFish tow cable length versus depth (cable with no fairing)

2.3 Layback and cable length for tow cable with 100 m hair fairing

The following graphs show the results for a tow cable with 100 m of hair fairing.

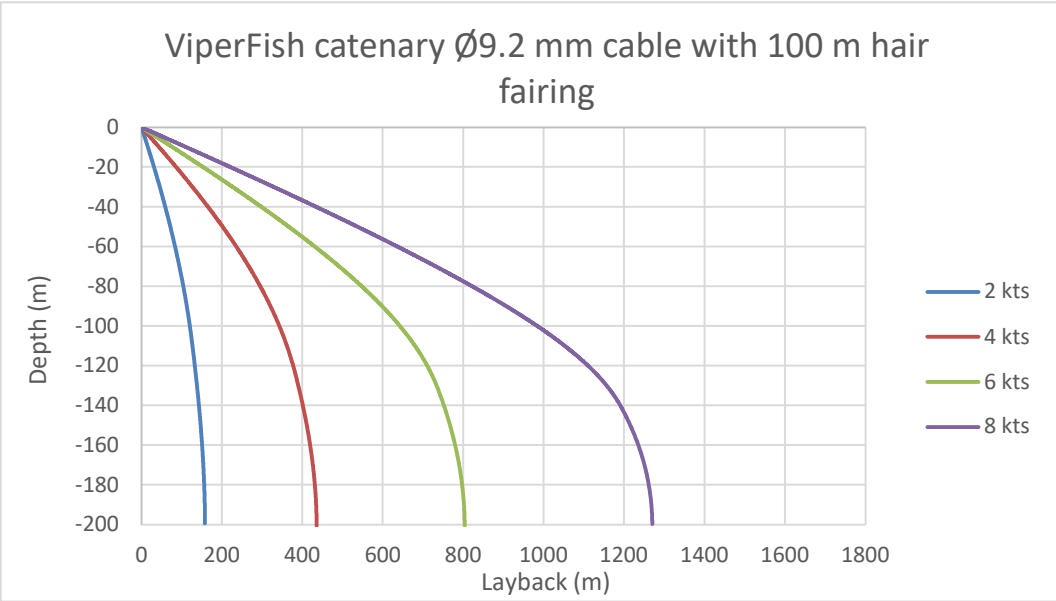

Figure 6 ViperFish cable layback after vessel versus depth (cable with 100 m hair fairing)

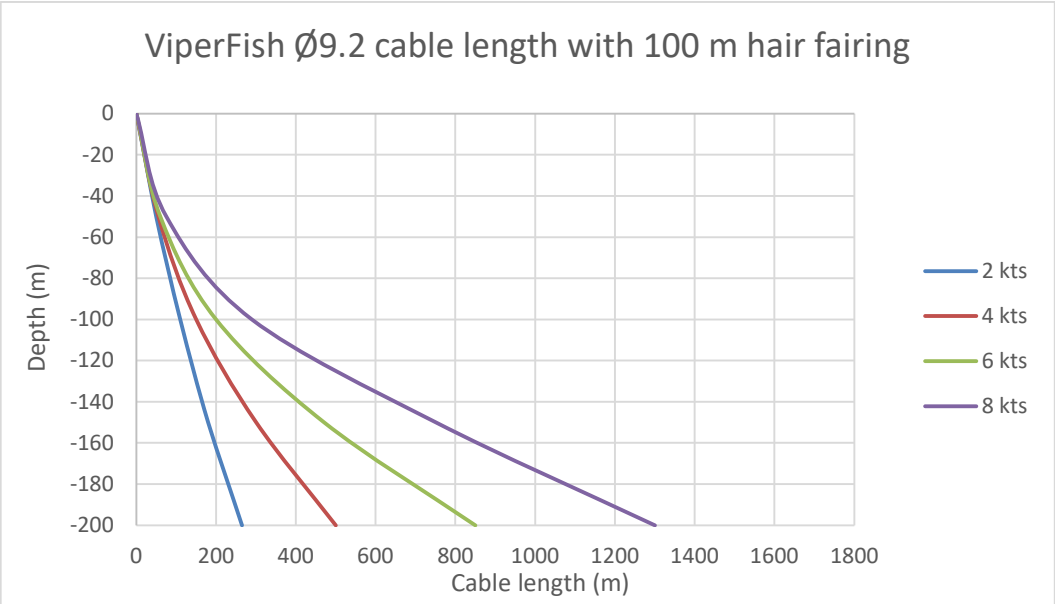

Figure 7 ViperFish tow cable length versus depth (cable with 100 m hair fairing)

2.4 Layback and cable length for tow cable with 500 m hair fairing (10 kts)

The following graphs show the results for a tow cable with 500 m of hair fairing.

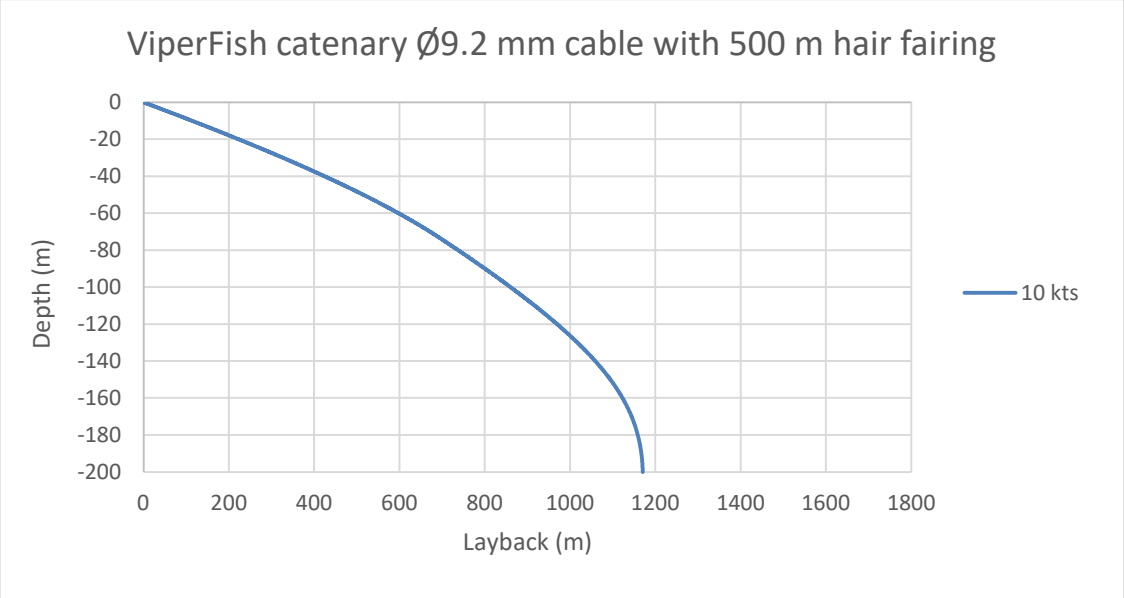

Figure 8 ViperFish cable layback after vessel versus depth (cable with 500 m hair fairing)

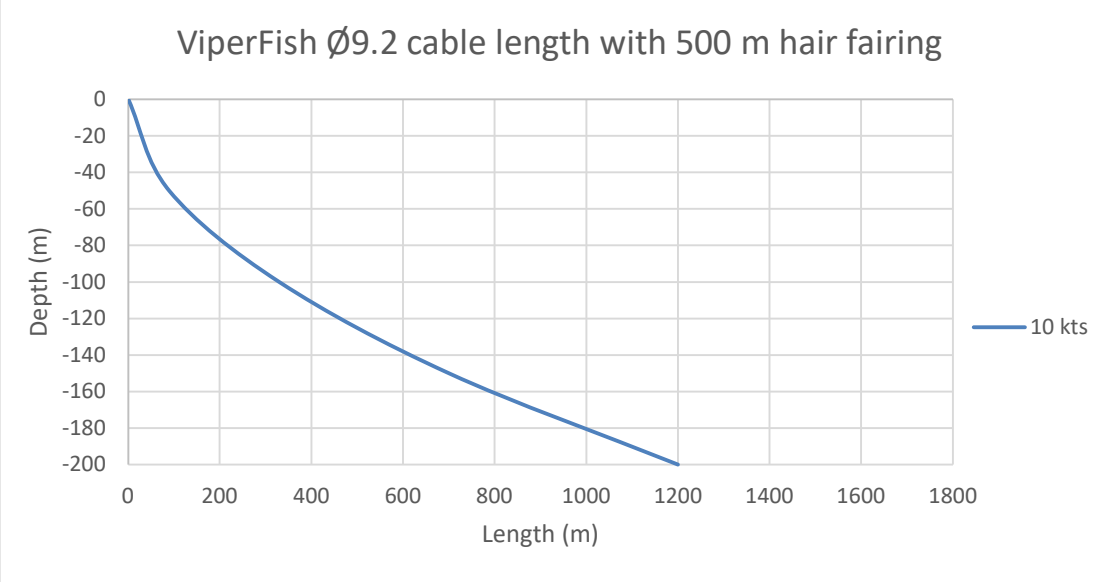

Figure 9 ViperFish tow cable length versus depth (cable with 500 m hair fairing)

3 Cable tension with ViperFish

When the cable with a ViperFish is towed through the water, there is a tension in the cable and the winch.

The tension is a result of a combination of factors: the depth of the cable, the STW of the ViperFish, the drag on the cable, the swept area of the cable and the fairing used on the cable.

The following graphs show the cable tension by depth at different velocities for cables with various lengths of hair fairing.

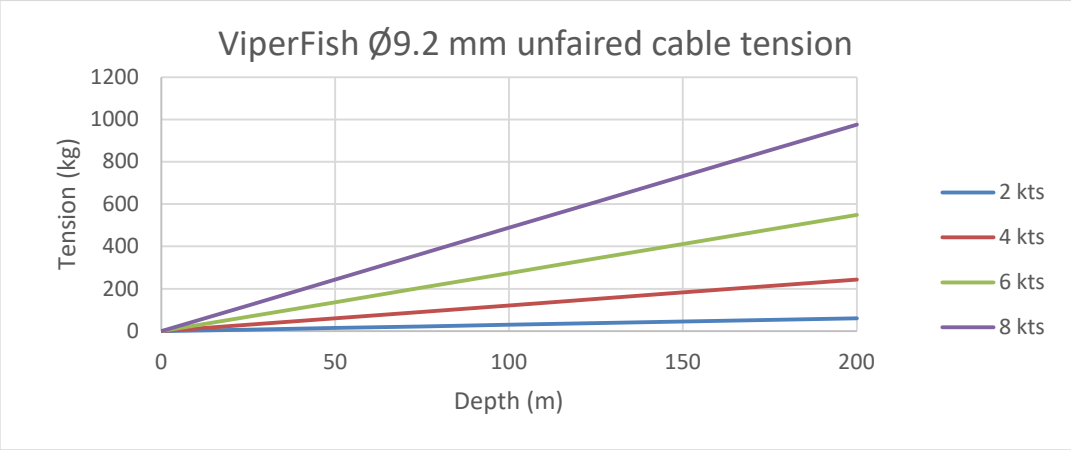

Figure 10 ViperFish tow cable tension versus depth at different velocities (cable with no fairing)

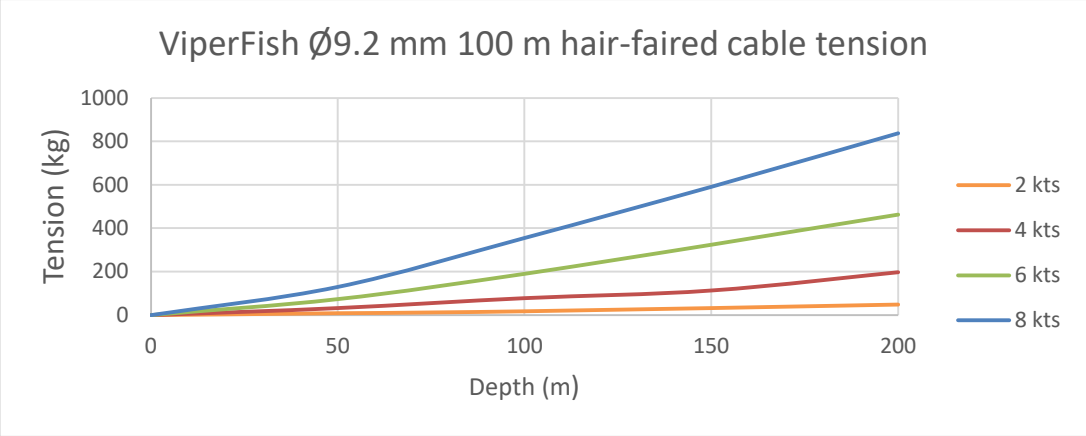

Figure 11 ViperFish tow cable tension versus depth at different speeds (cable with 100 m of fairing)

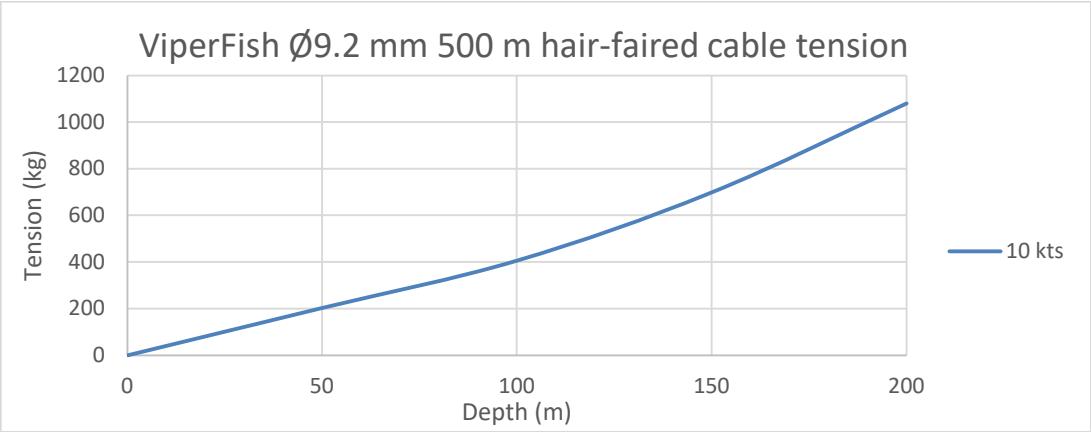

Figure 12 ViperFish tow cable tension versus depth (cable with 500 m of fairing)

4 Recommended cables

The following tow cables are recommended when towing a ViperFish:

- Hydrocable systems Type CLS9124-Model Ø9.2 mm fiber optic and 2+2 cores cable
- Hydrocable systems Type CS9124 Ø9.8 mm fiber optic and 2+2 cores cable

Please contact EIVA if you would like more information on these cables.